

ABM BUSINESS LOGIC BIBLE

§16 v2 — THE TEMPORAL INTELLIGENCE LAYER

10 Iterations · 10 Layers · 28 New Rules · 5 Integration Decisions · 1 Flagged Tension

§16 v1 consolidated every timer into one reference. §16 v2 reframes the layer itself. **§16 is not a schedule. It is the Temporal Intelligence Layer** — the dimension through which Signals (§3), Scoring (§5), Enrichment (§6), Outreach (§9) and Analytics (§13) all connect. In enterprise banking, a mediocre message at the right moment beats a brilliant message at the wrong one. The deepest discovery: the value of intelligence is a function of how fast it becomes action, not just how accurate it is — timing IS the competitive advantage, not merely the scheduling layer underneath one.

Iterations	10	Layers	10
New rules	28	Integration decisions	5 explicit

FIVE INTEGRATION DECISIONS + ONE FLAGGED TENSION

Opportunity Window modifies §5's Effective Opportunity scoring as a new input, not a parallel decay system alongside §6 v3's fact-freshness categories — window and freshness are different clocks measuring different things.

Temporal Escalation Events bind to GOV's existing Autonomy Ladder and the §3 EDGE catalogue — fast-tracking a CEO hire or SAMA mandate is an authority question already solved elsewhere, not a new bypass mechanism.

Signal decay SHAPE extends §6 v3's decay CATEGORIES (Operational/Tactical/Strategic/Structural get a shape — linear/exponential/step/plateau — layered on, not a competing decay model).

Initiative Progression Models are SIG-HYP hypotheses with a temporal dimension, carrying the same RCM stamp and anti-false-confidence discipline as any other hypothesis — never an unstamped oracle.

Competitive timing claims stay sourced-only, bound to §9 v2's existing Competitive Response discipline (SIG-VENDOR/SIG-NEWS) — §16 tracks Decimal's OWN speed; it never claims to know a competitor's actual response time without a sourced signal.

FLAGGED TENSION: 'outreach before the opportunity fully materializes' (Iteration 9) is in real tension with §14 v2's OPP-LIFE-01 birth conditions — resolved by naming it explicitly as Hypothesis-driven outreach, not Opportunity-driven outreach. See Layer 9.

THE TEN LAYERS — §16 v2 ARCHITECTURE MAP

#	Layer	From iter.	What it adds
1	Clock Collision (Sched)	1	Independently-correct timers can still spike infrastructure if they fire together
2	Latency Chain (Lat)	2	End-to-end Reality-to-Action latency, not the sum of locally-optimal component timers
3	Timing Priority (Class)	3	Real-Time/Operational/Strategic/Governance — not all breaches equally urgent
4	Human Clock (Hum)	4	Human SLAs are probabilistic (Ramadan, travel, quarter-end), not deterministic
5	Timing Economics (Econ)	5	Every timer is a freshness-vs-cost tradeoff requiring stated justification
6	Opportunity Window (Opp)	6	Window ≠ freshness; modifies §5 Effective Opportunity, doesn't replace §6 v3 decay
7	Temporal Dependency (Dep)	7	Fast-track via GOV/EDGE, not a new bypass mechanism
8	Signal Decay Shape (Decay)	8	Linear/exponential/step/plateau layered onto §6 v3's existing categories
9	Strategic Timing Intelligence (Strat)	9	Initiative Progression as a stamped SIG-HYP hypothesis; flagged tension w/ §14 v2
10	Time as Competitive Advantage (Adv)	10	Sourced-only competitive framing; Decimal's own speed is the real metric

LAYER 1 — CLOCK COLLISION (SCHED)

Monday 09:00: CRM pull, signal poll, enrichment refresh, analytics job, and retry-queue sweep can all fire simultaneously. Each timer is individually correct; the infrastructure still sees one spike. §16 v1 listed every timer; it never asked whether they collide.

Rule ID	Phase	Specification
TIME-SCHED-01	P1	A System Clock Map visualizes every recurring timer in §16.0–16.8 on one timeline, making simultaneous-fire clusters visible at a glance — built from the existing §16 v1 matrix, not a new timer inventory.
TIME-SCHED-02	P1	New recurring timers are deliberately offset from existing ones at the SAME cadence to avoid clustering (e.g. if Loop 1 and Loop 4 both run 'every 30 days,' they are scheduled on different days of that cycle, not the same hour of the same day).
TIME-SCHED-03	P2	Timer design considers system load (API rate limits, compute cost, concurrent-job ceiling) alongside business need — a timer change proposal (per §16 v1 TIME-MATRIX-05's maintenance obligation) must show it doesn't create a new collision before approval.

LAYER 2 — LATENCY CHAIN (LAT)

Rule ID	Phase	Specification
TIME-LAT-01	P1	Every critical workflow (signal→enrichment→scoring→outreach) receives an explicit End-to-End Response SLA: e.g. Executive-move signal → first outreach ≤ 48 hours, summing across every component timer in the chain (§16.0–16.3), not just each stage's local SLA.
TIME-LAT-02	P1	§13 v2 tracks Reality-to-Action Latency as its own KPI — distinct from any single component's performance — because a chain of five individually-fast stages can still produce an unacceptably slow end-to-end response if their handoffs aren't coordinated.

LAYER 3 — TIMING PRIORITY (CLASS)

Rule ID	Phase	Specification
TIME-CLASS-01	P1	Every timer in §16.0–16.8 is classified Real-Time (engagement webhooks, reply detection), Operational (CRM pulls, signal polling), Strategic (enrichment refresh, opportunity-window tracking), or Governance (Learning Review Board, Resilience Investment review).
TIME-CLASS-02	P1	A breached timer's response priority is set by its Class, not by raw count of breaches — ten missed Governance-class reviews in a quarter is less urgent than one missed Real-Time reply-detection window, even though the latter is a single incident.

LAYER 4 — HUMAN CLOCK (HUM)

Rule ID	Phase	Specification
TIME-HUM-01	P2	Human-facing SLAs (§16.5: review SLA, handoff SLA, escalation SLA) carry a Capacity Context — Ramadan, conference weeks, quarter-end, individual leave calendars — sourced from the same KSA-calendar constraints §17 will formally specify, plus any team-level capacity data already available.
TIME-HUM-02	P1	Human SLA calculations adjust for Available Capacity, not clock time alone: a 4-hour review SLA during a known low-capacity period (Ramadan reduced hours, a public holiday) is interpreted against capacity-adjusted business hours, not literal wall-clock hours — this does NOT relax the SLA's intent, it correctly accounts for when 'business hours' actually exist.

LAYER 5 — TIMING ECONOMICS (ECON)

Every timer is a tradeoff between freshness and cost — polling every minute maximizes freshness and maximizes cost/complexity; polling daily minimizes both. §16 v1 listed durations without ever asking why THAT duration specifically.

Rule ID	Phase	Specification
TIME-ECON-01	P2	Every timer's specification (going forward; not retroactively re-justifying all 60+ existing ones) requires an Economic Justification — why this frequency, why not faster, why not slower — recorded alongside the timer in §16.
TIME-ECON-02	P2	Timer reviews (part of §16 v1 TIME-MATRIX-05's maintenance obligation) evaluate Value Generated vs Cost Consumed using the SAME Expected-Value family as §6 v3/§15 v2 (marginal-value gating, Business Impact estimation) — not a fourth independent value-estimation method for timing specifically.
TIME-ECON-03	P1	A timer frequency change requires business justification submitted as an A4 governance proposal (GOV-MOD-01), exactly like any other learned or proposed system change — 'it felt right' is not sufficient grounds to change a cadence that affects cost or collision risk (Layer 1).

LAYER 6 — OPPORTUNITY WINDOW THEORY (OPP)

A signal can remain technically FRESH (per §3/§6 v3's decay model) while the actual buying window has already closed — 90 days after a SAMA regulatory release, vendors are shortlisted regardless of how fresh the original signal still reads. Freshness and Opportunity Window are different clocks.

Rule ID	Phase	Specification
TIME-OPP-01	P2	Every signal category receives an Opportunity Window Profile — the typical duration the underlying BUYING OPPORTUNITY (not the signal's freshness) stays open: Leadership Hire 90–180 days; Regulatory Change 30–90 days; Funding Event 60–120 days; RFP 15–45 days. This is a NEW clock, separate from §3/§6 v3's signal-freshness decay.
TIME-OPP-02	P1	§5's Effective Opportunity scoring (§1/§5 v3) takes Opportunity Window Remaining as an ADDITIONAL multiplier input alongside Signal Strength — a strong signal deep into its window's close scores differently than the same strength signal early in its window. This modifies the EXISTING formula; it does not create a parallel scoring path.
TIME-OPP-03	P1	An account approaching its Opportunity Window's expiry receives a priority boost in §5 v3's portfolio-ranking (TIER-PORT-002) specifically because of urgency — a closing window is itself a reason to act sooner, distinct from and additive to the account's raw Effective Opportunity value.

LAYER 7 — TEMPORAL DEPENDENCY THEORY (DEP)

A board appointment or SAMA mandate may justify skipping the standard signal→enrichment→scoring→review sequence's normal pacing entirely. This is an authority question — GOV and the §3 EDGE catalogue already answer 'who gets to act fast and on whose authority'; §16 v2 binds the timing dimension to that existing answer rather than inventing a second bypass system.

Rule ID	Phase	Specification
TIME-DEP-01	P1	Named Temporal Escalation Events (CEO/CTO hire, strategic acquisition, major RFP, SAMA mandate, Vision 2030 initiative announcement) are recognized as a SUBSET of §3's existing P1_CRITICAL signal classification, cross-referenced against GOV's Autonomy Ladder (which tier of action they unlock) and the EDGE catalogue (which scenario, if any, they also trigger) — not a new classification system.
TIME-DEP-02	P1	A recognized Temporal Escalation Event uses the FAST-TRACK processing path already implied by P1_CRITICAL's existing 1-hour score-recalculation / 2-hour enrichment-trigger SLA (§3 §3.4) — §16 v2 makes explicit that this IS the fast-track path, rather than describing a separate queue that doesn't exist.
TIME-DEP-03	P2	Variable process speed (some workflows compress, most don't) is bounded by GOV: a Temporal Escalation Event can accelerate TIMING but never bypass an AUTHORITY gate (e.g. it cannot skip QC-RULE-009's permanent political-sensitivity human review just because it's time-urgent) — speed and authority are independent dimensions; this rule keeps them that way.

LAYER 8 — SIGNAL DECAY SHAPE (DECAY)

An executive resignation holds high value, high value, high value, then drops to near-zero the moment a successor is named — a step function, not a smooth curve. A digital-transformation announcement decays gradually. §6 v3 already gave facts category-based decay DURATIONS; this layer adds decay SHAPE.

Rule ID	Phase	Specification
TIME-DECAY-01	P2	Every signal category receives a Decay Function Type — Linear, Exponential, Step-function, or Plateau — layered ONTO §6 v3 ENR-DECAY-01's existing Operational/Tactical/Strategic/Structural duration categories, not a replacement for them. Example: LEADERSHIP_HIRE (executive departure sub-type) = Step-function (high value until a successor is named, then near-zero); DIGITAL_INITIATIVE = gradual Exponential decay.
TIME-DECAY-02	P1	§4/§5 scoring applies the signal-specific Decay Function Type, not a universal smooth curve, when computing freshness-weighted Signal Strength (§4 v1 SCORE-SIG-01) — a step-function signal's value should not be modeled as gradually fading when its real behavior is a cliff.
TIME-DECAY-03	P2	§13 v2's Model Drift mechanism (ANL-DRIFT-01/02) periodically validates decay-shape assumptions against actual outcome data — a signal category whose assumed decay shape no longer matches observed value-over-time is itself a drift signal, using the SAME Predictive Stability Score mechanism, not a new validation method specific to decay curves.

LAYER 9 — STRATEGIC TIMING INTELLIGENCE (STRAT)

Cloud migration often precedes core modernization, which often precedes an API layer initiative, which often precedes a partner-ecosystem opportunity. Predicting the NEXT opportunity in a known sequence — not just reacting to the current one — is a temporal hypothesis, and must carry the same epistemic discipline as any other hypothesis in this Bible.

Rule ID	Phase	Specification
TIME-STRAT-01	P2	Initiative Progression Models (e.g. Cloud → API → Partner → Marketplace) are specified and maintained as named patterns, derived from observed account histories (Tier 6 historical data) — not invented from intuition.
TIME-STRAT-02	P1	INTEGRATION DECISION: a predicted future opportunity from an Initiative Progression Model is created as a SIG-HYP hypothesis (§3 reasoning streams) with a full EPIS-RCM stamp — NOT an unstamped prediction. Its confidence is explicitly falsifiable (EPIS-RCM-03) and it is never presented to a human as a confirmed future event.
TIME-STRAT-03	P1	FLAGGED TENSION, RESOLVED: 'outreach may begin before the opportunity fully materializes' is permitted ONLY as explicitly-labeled Hypothesis-driven outreach — it does NOT satisfy or bypass §14 v2 OPP-LIFE-01's three-condition Opportunity birth test. An Opportunity entity is not created merely because a Progression Model predicts one is likely; outreach grounded in a SIG-HYP hypothesis is clearly distinguished (in the dossier, in QC, in the handoff package) from outreach grounded in a confirmed Opportunity, so a human never mistakes a good guess for a qualified deal.

LAYER 10 — TIME AS COMPETITIVE ADVANTAGE (ADV)

The deepest discovery. Two firms with identical intelligence about the same signal produce different outcomes purely based on response speed. The value of intelligence is a function of how fast it becomes action — but claiming a competitive speed advantage requires actual evidence about competitors, which this Bible has always required to be sourced, never assumed.

Rule ID	Phase	Specification
TIME-ADV-01	P1	Signal-to-Action Time is tracked for every opportunity — Decimal's OWN elapsed time from signal detection to first qualified outreach — as a standing §13 v2 KPI, feeding the Layer 2 End-to-End Latency metric.
TIME-ADV-02	P1	INTEGRATION DECISION: any claim about COMPETITOR response speed is bound to §9 v2 OUT-COMP-01's sourced-only discipline (SIG-VENDOR/SIG-NEWS evidence only) — §16 v2 never assumes or estimates a competitor's speed without a sourced signal. Decimal's own Signal-to-Action Time, measured directly, is the primary and always-available metric; competitive comparison is a sometimes-available secondary one.
TIME-ADV-03	P2	§16's stated purpose, as the Temporal Intelligence Layer, is to convert intelligence into action faster — subject to every other governance constraint already specified (GOV authority gates, §8 QC hard gates, §13 v2 Constitutional Constraints) — speed is optimized WITHIN those bounds, never traded against them.

§16 v2 RULE ROLL-UP & PLACEMENT STATUS

Layer	Rules	Phase	Cross-section binding
1 Sched	TIME-SCHED-01..03	P1 / P2	§16 v1's existing timer inventory (visualized, not re-collected)
2 Lat	TIME-LAT-01..02	P1	§13 v2 new KPI
3 Class	TIME-CLASS-01..02	P1	§16 v1's full timer table (classified, not re-specified)
4 Hum	TIME-HUM-01..02	P1 / P2	§17 KSA calendar (pending), §16.5 SLA table
5 Econ	TIME-ECON-01..03	P1 / P2	§6 v3/§15 v2 Expected-Value family, GOV-MOD-01
6 Opp	TIME-OPP-01..03	P1 / P2	§5/§5 v3 Effective Opportunity (modified input), §6 v3 (distinct from)
7 Dep	TIME-DEP-01..03	P1 / P2	§3 P1_CRITICAL, GOV Autonomy Ladder, EDGE catalogue
8 Decay	TIME-DECAY-01..03	P1 / P2	§6 v3 ENR-DECAY-01 (extended), §13 v2 ANL-DRIFT
9 Strat	TIME-STRAT-01..03	P1 / P2	SIG-HYP, EPIS-RCM-03, §14 v2 OPP-LIFE-01 (tension resolved)
10 Adv	TIME-ADV-01..03	P1 / P2	§9 v2 OUT-COMP-01 (sourced-only), §13 v2 KPI

§16 v2 COMPLETE — WHAT IT ESTABLISHED

§16 is no longer a reference table. It is the **Temporal Intelligence Layer** connecting Signals, Scoring, Enrichment, Outreach and Analytics through the dimension of time: it visualizes clock collisions before they spike infrastructure, measures end-to-end Reality-to-Action latency rather than only local timer performance, classifies every timer's urgency by business function, accounts for human time as probabilistic rather than deterministic, requires economic justification for every cadence, separates opportunity-window urgency from signal freshness as two distinct clocks feeding §5's scoring, binds fast-tracking to authority gates that already exist rather than inventing new ones, gives decay a shape as well as a duration, treats predicted future opportunities as stamped hypotheses rather than confident forecasts, and — the deepest discovery — recognizes that the value of intelligence is a function of how fast it becomes action, measured honestly against Decimal's own speed rather than assumed competitor inferiority. 28 new rules, five integration decisions, one explicitly flagged and resolved tension with §14 v2's Opportunity birth conditions.

PLACEMENT MAP UPDATE

§16 status remains Complete — all ten layers resolved to concrete rules bound to existing architecture; no new undecided open questions (the one real tension found, Layer 9, was resolved within this document rather than left open).

ON THE HORIZON

§17 KSA-Specific Business Rules — the Bible's last 'Not started' section, and per consensus across every recent review, likely the richest remaining territory. §16 v2's Layer 4 (Human Clock) and Layer 7 (Temporal Dependency) both explicitly defer their calendar-specific content to §17; building §17 will retroactively complete both.